



Lab Assignment/Task Report

Only for course Teacher					
	Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage	25%	50%	75%	100%	
Clarity					
Content Quality					
Spelling & Grammar					
Organization and Formatting					
Total obtained mark					
Comments					

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Batch: 46th

Section: A

Course Code: CSE 124

Course Name: [Object Oriented Programming Lab](#)

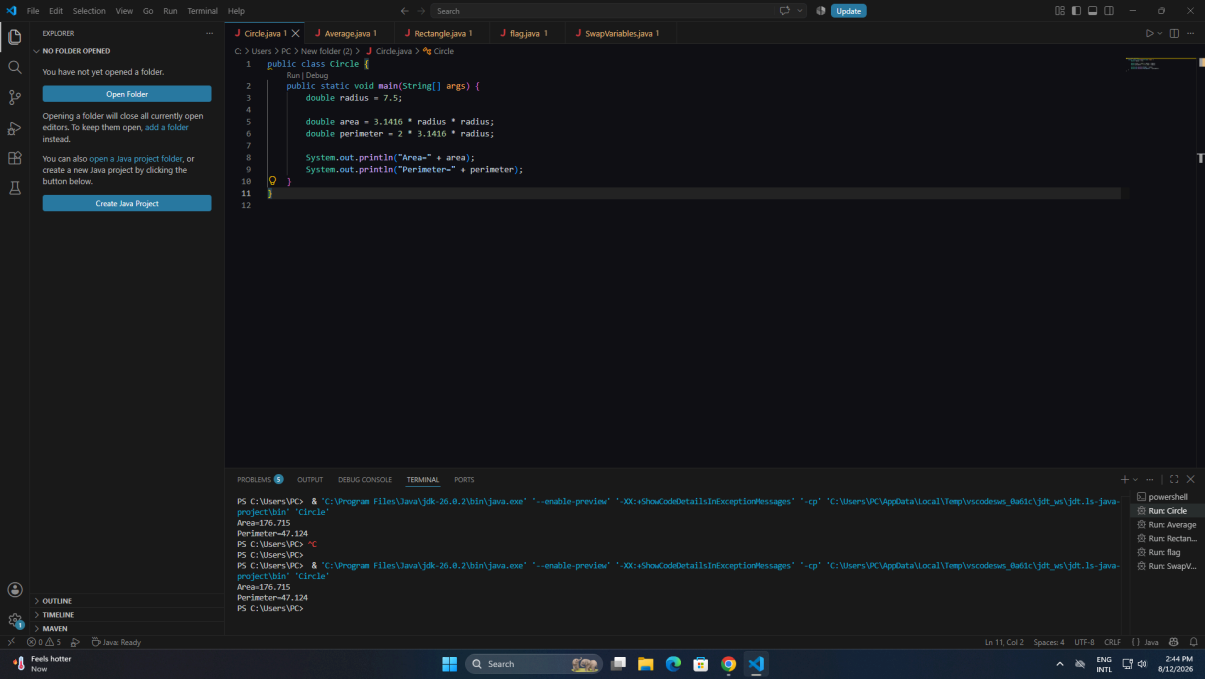
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Designation: Lecturer

Submission Date: 29/07/2026

11. Circle: Area and Perimeter

Question: Write a Java program to print the area and perimeter of a circle.



The screenshot shows a Java IDE with a project named 'Circle'. The main editor displays the following Java code:

```
1 public class Circle {  
2     public static void main(String[] args) {  
3         double radius = 7.5;  
4  
5         double area = 3.1416 * radius * radius;  
6         double perimeter = 2 * 3.1416 * radius;  
7  
8         System.out.println("Area=" + area);  
9         System.out.println("Perimeter=" + perimeter);  
10    }  
11 }  
12
```

The bottom panel shows the terminal output:

```
PS C:\Users\PC> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' --enable-preview --XX:+ShowCodeDetailsInExceptionMessages -cp 'C:\Users\PC\AppData\Local\Temp\vscodewas_8u61c\jdt_ws\jdt.ls-java-project\bin' "Circle"  
Area=176.715  
Perimeter=47.124  
PS C:\Users\PC>  
PS C:\Users\PC> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' --enable-preview --XX:+ShowCodeDetailsInExceptionMessages -cp 'C:\Users\PC\AppData\Local\Temp\vscodewas_8u61c\jdt_ws\jdt.ls-java-project\bin' "Circle"  
Area=176.715  
Perimeter=47.124  
PS C:\Users\PC>
```

Definition: This program calculates the area and perimeter of a circle.

12. Average of Three Numbers

Question: Write a Java program that takes three numbers as input to calculate and print the average of the numbers.

The screenshot shows the Visual Studio Code editor with a Java file named `Average.java`. The code defines a `public class Average` with a `main` method that takes three arguments, calculates their average, and prints it. The terminal output shows the program running successfully and printing `Average=20.0`.

```
1 public class Average {
2     public static void main(String[] args) {
3         double a = 10;
4         double b = 20;
5         double c = 30;
6
7         double average = (a + b + c) / 3;
8
9         System.out.println("Average=" + average);
10    }
11 }
12
```

```
PS C:\Users\PC> & "C:\Program Files\Java\jdk-26.0.2\bin\java.exe" --enable-preview "-XX:+ShowCodeDetailsInExceptionMessages" -cp "C:\Users\PC\AppData\Local\Temp\vscodewsg_8u61c\jdt_ws\jdt.ls-java-project\bin" Average
Average=20.0
PS C:\Users\PC> ^C
PS C:\Users\PC> & "C:\Program Files\Java\jdk-26.0.2\bin\java.exe" --enable-preview "-XX:+ShowCodeDetailsInExceptionMessages" -cp "C:\Users\PC\AppData\Local\Temp\vscodewsg_8u61c\jdt_ws\jdt.ls-java-project\bin" Average
Average=20.0
PS C:\Users\PC>
```

Definition: This program calculates the average of three numbers.

13. Rectangle: Area and Perimeter

Question: Write a Java program to print the area and perimeter of a rectangle.

The screenshot shows the Visual Studio Code editor with a Java file named `Rectangle.java`. The code defines a `public class Rectangle` with a `main` method that calculates the area and perimeter of a rectangle and prints them. The terminal output shows the program running successfully and printing `Area=46.75` and `Perimeter=28.0`.

```
1 public class Rectangle {
2     public static void main(String[] args) {
3         double width = 5.5;
4         double height = 8.5;
5
6         double area = width * height;
7         double perimeter = 2 * (width + height);
8
9         System.out.println("Area=" + area);
10        System.out.println("Perimeter=" + perimeter);
11    }
12 }
```

```
PS C:\Users\PC> & "C:\Program Files\Java\jdk-26.0.2\bin\java.exe" --enable-preview "-XX:+ShowCodeDetailsInExceptionMessages" -cp "C:\Users\PC\AppData\Local\Temp\vscodewsg_8u61c\jdt_ws\jdt.ls-java-project\bin" Rectangle
Area=46.75
Perimeter=28.0
PS C:\Users\PC> ^C
PS C:\Users\PC> & "C:\Program Files\Java\jdk-26.0.2\bin\java.exe" --enable-preview "-XX:+ShowCodeDetailsInExceptionMessages" -cp "C:\Users\PC\AppData\Local\Temp\vscodewsg_8u61c\jdt_ws\jdt.ls-java-project\bin" Rectangle
Area=46.75
Perimeter=28.0
PS C:\Users\PC>
```

Definition: This program calculates the area and perimeter of a rectangle.

14. American Flag Display

Question: Write a Java program to print an American flag on the screen.

The screenshot shows an IDE with a Java file named `flag.java`. The code defines a `public class flag` with a `main` method that prints a flag pattern using stars and equal signs. The terminal output shows the execution of the program, which prints a flag pattern.

```
1 public class flag {
2     public static void main(String[] args) {
3         System.out.println("*****");
4         System.out.println("*****");
5         System.out.println("*****");
6         System.out.println("*****");
7         System.out.println("*****");
8         System.out.println("*****");
9         System.out.println("*****");
10        System.out.println("*****");
11        System.out.println("*****");
12    }
13 }
14
```

The terminal output shows the execution of the program, which prints a flag pattern.

```
PS C:\Users\PC>
PS C:\Users\PC> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' --enable-preview --XX:ShowCodeDetailsInExceptionMessages -cp 'C:\Users\PC\AppData\Local\Temp\vscodewse_0u61c\jdt_ws\jdt.ls-java-project\bin' *.flag
*****
*****
*****
*****
*****
*****
*****
*****
*****
*****
PS C:\Users\PC>
```

Definition: This program prints an American flag using stars and equal signs.

15. Swap Variables

Question: Write a Java program to swap two variables.

The screenshot shows an IDE with a Java file named `SwapVariables.java`. The code defines a `public class SwapVariables` with a `main` method that swaps two variables, `a` and `b`, using a temporary variable `temp`. The terminal output shows the execution of the program, which prints the swapped values of `a` and `b`.

```
1 public class SwapVariables {
2     public static void main(String[] args) {
3         int a = 10;
4         int b = 20;
5
6         int temp = a;
7         a = b;
8         b = temp;
9
10        System.out.println("a=" + a);
11        System.out.println("b=" + b);
12    }
13 }
```

The terminal output shows the execution of the program, which prints the swapped values of `a` and `b`.

```
PS C:\Users\PC> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' --enable-preview --XX:ShowCodeDetailsInExceptionMessages -cp 'C:\Users\PC\AppData\Local\Temp\vscodewse_0u61c\jdt_ws\jdt.ls-java-project\bin' *.SwapVariables
a=20
b=10
PS C:\Users\PC>
PS C:\Users\PC> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' --enable-preview --XX:ShowCodeDetailsInExceptionMessages -cp 'C:\Users\PC\AppData\Local\Temp\vscodewse_0u61c\jdt_ws\jdt.ls-java-project\bin' *.SwapVariables
```

Definition: This program swaps the values of two variables.